Hemodynamic impact of an off-the-shelf thoracoabdominal multi-branched endoprosthesis (TAMBE): a patient-specific CFD pilot study

Ethan Farah[1], Allison Marsden[2], Jason T. Lee[3], and Kenneth Tran[3]

[1]Department of Mechanical Engineering, Stanford University

[2]Department of Pediatric Cardiology, Stanford University

[3]Division of Vascular Surgery, Stanford University

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Abstract

Introduction The recently approved thoracoabdominal multi-branched endoprosthesis (TAMBE) is the first commercially available off-the-shelf device in the U.S. for treating thoracoabdominal aortic pathology. This pilot study aims to evaluate the hemodynamic performance of the TAMBE device using patient-specific computational fluid dynamics (CFD) modeling.

Methods Retrospective analysis was conducted on 10 consecutive patients undergoing elective thoracoabdominal aortic aneurysm (TAAA) repair with the Gore TAMBE device. Pre- and post-operative thoracoabdominal anatomy was segmented from CT angiograms to generate three-

dimensional volume meshes using SimVascular. Patient-specific, allometrically scaled pulsatile inflow conditions were applied. Outlet boundary conditions were modeled using three-element Windkessel representations. Rigid-wall, pulsatile flow simulations were performed for pre- and post-operative models under identical inflow and outflow conditions. Computationally derived pressure, flow rates, and wall shear stress (WSS) for renal and visceral branches were compared between pre- and post-operative states using paired t-tests.

Results TAMBE implantation resulted in a modest reduction in computationally estimated peak renal artery flow rates (15.55 ± 3.00 vs. 15.04 ± 3.37 ml/s, $p = .046$) and peak renal artery pressure (113.1 ± 8.9 vs. 108.7 ± 8.5 mmHg, $p = .026$). Most stented renal arteries (75%) experienced less than a 10% change in hemodynamics. However, five stented renal arteries showed reductions of 13–22% in peak pressure or flow. Exploratory geometric analysis revealed that these reductions were most strongly associated with the combination of branch tortuosity and distal angulation relative to the aortic axis, consistent with a path-complexity mechanism rather than any single angular measurement. No significant hemodynamic changes were observed in the celiac or superior mesenteric arteries. The fractional area of abnormal wall shear stress (WSS < 10 or > 70 dyn/cm^2) also remained unchanged post-operatively.

Conclusions In this pilot cohort, TAMBE implantation was associated with largely preserved renovisceral hemodynamics and only a modest reduction in peak renal perfusion metrics. No branch occlusions or clinically significant stenoses were identified on available post-operative imaging during early follow-up. Larger cohort studies with duplex ultrasound validation and longer clinical follow-up will be needed to confirm these findings and to evaluate whether multivariate branch-geometry assessment can improve patient selection and configuration planning.

Keywords thoracoabdominal aneurysm; TAMBE; branched endograft; computational fluid dynamics; Windkessel; renal artery; wall shear stress

Introduction

Over the past decade, fenestrated/branched endovascular repair (F/BEVAR) has expanded treatment to patients unfit for open surgery and is widely used with favorable early outcomes, although durability and reintervention needs remain important considerations [1]. Contemporary reviews describe high technical success but significant rates of reintervention, underscoring the need to better understand device–flow interactions in the renovisceral segment [2, 3].

In January 2024, the U.S. FDA approved the GORE EXCLUDER Thoracoabdominal Branch Endoprosthesis (TAMBE) device, the first FDA-approved, off-the-shelf branched solution for aneurysmal disease involving the visceral aorta. This device was quickly adopted at many surgical centers, leading to a new treatment paradigm for both elective and emergent thoracoabdominal pathology. The device design is unique compared to other commercially available branch devices outside the United States, namely with the branch design resulting in significant "proximalization" of the renovisceral branches. This results in significantly longer lengths of stented branch vessels compared to other branched devices or fenestrated strategies. Indeed, the latest clinical data suggest that renal branches carry a disproportionate share of target vessel events after off-the-shelf multibranched repair. In the pivotal TAMBE trial, approximately 15% of patients had at least one branch occlusion by 1 year, and vessel-level freedom from target vessel instability was lower in renals compared to mesenteric branches [4]. Additional series report that renal arteries are more prone to patency loss, with risk influenced by smaller renal diameters and device/branch configuration [5–7].

Patient-specific computational fluid dynamics (CFD) modeling is a powerful emerging tool to analyze the hemodynamic effect of varying aortic graft configurations. Previous studies from our group have demonstrated that local adverse metrics such as abnormal wall shear stress (WSS) and residence time can portend future thrombotic risk [16–19]. Given these data, we hypothesized that patient-specific CFD may reveal potential adverse hemodynamic changes associated with TAMBE implantation.

Methods

Study design and cohort

This study was a single-center, retrospective cohort of consecutive patients undergoing elective thoracoabdominal aortic aneurysm using the TAMBE device. We included the first 10 patients at our institution treated with this device in consecutive fashion. Additional patients were not included due to computational time constraints. The study was approved by our local institutional review board and reported according to the STROBE guideline for observational studies.

Patients and imaging

Table 1: Baseline cohort characteristics ($n = 10$). Values are median [IQR] or n (%). Abbreviations: CAD, coronary artery disease; CHF, congestive heart failure; CKD, chronic kidney disease; SVS, Society for Vascular Surgery; IQR, interquartile range. Crawford extent reported as proportion of thoracoabdominal aortic aneurysm (TAAA) types III–V. Aortic maximum diameter measured on pre-operative CTA.

	$n = 10$
Age	74 [69–78]
Female	2 (20%)
CAD	6 (60%)
CHF	3 (30%)
Hypertension	10 (100%)
Diabetes	0 (0%)
CKD	3 (30%)
Tobacco History	10 (100%)
SVS Co-Morbidity Score	10.3 [8–14]
Crawford Extent	
III	3 (30%)
IV	6 (60%)
V	1 (10%)
Aortic maximum diameter (mm)	66 [55–88]

Among 10 patients, the median age was 74 [69–78] years and 20% were female. Comorbidities were common: hypertension 100%, CAD 60%, CHF 30%, and CKD 30%; tobacco history 100%. TAAA extents were Crawford III 30%, IV 60%, and V 10%, and the median maximum aortic

diameter was 66 mm [55–88]. The first post-operative CTA performed within one month of the operation was used for CFD analysis.

Geometry reconstruction

Pre-operative and first post-operative contrast-enhanced CT angiograms were imported into SimVascular (open-source; version 2023.05.23). The modeling workflow followed the standard SimVascular pipeline of plotting vessel paths, segmenting the CT, generating a model and mesh, and finally simulating heart beats [9, 10].

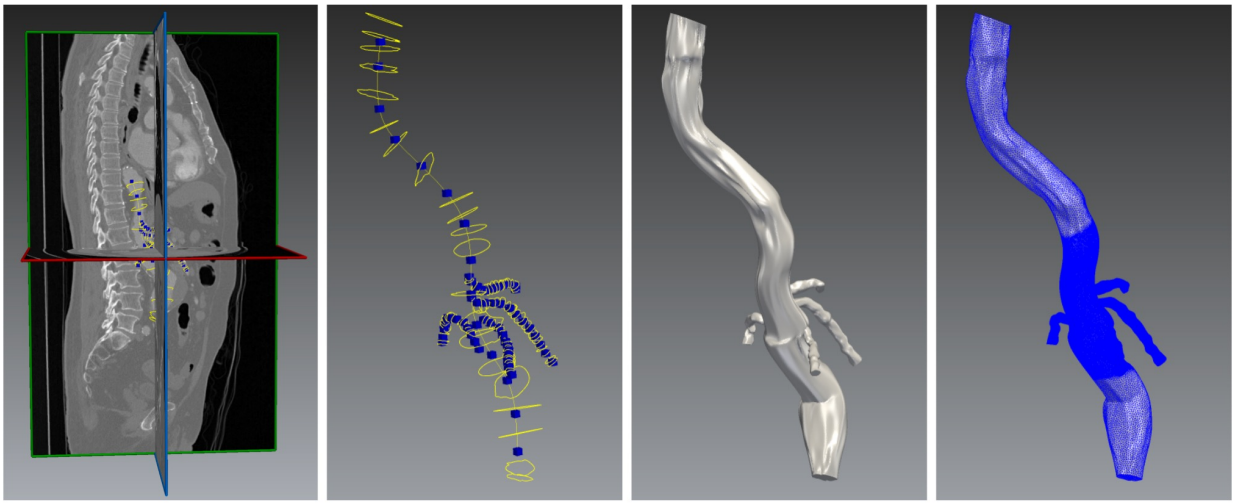
A centerline path for the thoracoabdominal aorta was drawn from the proximal descending thoracic aorta just distal to the left subclavian artery to approximately 10 mm above the aorto-iliac bifurcation. Separate paths were created for each renovisceral branch of interest—celiac trunk, superior mesenteric artery (SMA), left renal, and right renal—with control points spaced approximately 10 to 20 mm apart and Fourier smoothing (mode = 10) applied to remove high-frequency path jitter while preserving curvature.

Cross-sectional 2D segmentations were performed on planes orthogonal to each smoothed path at about 10 to 20 mm station spacing and then lofted to create branch surfaces. We initially used free-form contours (to closely match lumen shape) but found this introduced extraneous local undulations that produced poor element quality; therefore, contours were regularized to circles/ellipses while preserving cross-sectional area. All inlets/outlets were planar capped and named (aorta, celiac, SMA, left, right) for later boundary-condition assignment.

Pre- and post-operative models were not rigidly co-registered; instead, we standardized boundary placement using the same anatomical landmarks (proximal path start just distal to the left subclavian; distal stop 10 mm above the aorto-iliac bifurcation) and kept identical boundary conditions across pre/post models to isolate geometric effects.

Meshing

A tetrahedral volume mesh was generated using SimVascular (TetGen) with local size controls at surfaces and caps to control element density in regions of interest, based on previous mesh convergence studies [16]. A coarse mesh was used in the thoracic and distal aorta to minimize computational time (Fig. 1).



(a) (b) (c) (d)

Figure 1: Example of creating a mesh from a CT scan. (a) the CT scan of a patient with the selected arteries pathed (indicated by the "spine" connected by blue cubes) and segmented (indicated by the concentric yellow splines); (b) the same pathed and segmented arteries without the CT scan background; (c) the same arteries roughly modeled; (d) a fine mesh of the same arteries with coarse regions (indicated by a low blue density) along walls of the aorta.

Boundary conditions

Inflow (AAA reference waveform + scaling)

At the thoracic aortic inlet we prescribed an estimated patient-specific pulsatile flow waveform using the supraceliac (SC) and infrarenal (IR) ensemble-average waveforms reported by Les et al. [8]. For each subject, we scaled the reference waveform amplitude by body-size (BSA) allometry and

paced it by the patient's heart rate. Mean branch flows were allocated from the scaled inlet using fixed fractions (α) chosen *a priori* from clinical physiology and our prior work: $\alpha_{\text{celiac}} = 0.12$, $\alpha_{\text{SMA}} = 0.12$, $\alpha_{\text{left-renal}} = 0.14$, $\alpha_{\text{right-renal}} = 0.14$, $\alpha_{\text{distal-aorta}} = 0.48$ (sum $= 1.0$). Identical inflow was used for each patient's pre- and post-operative models to isolate geometric effects [8, 9, 13].

Outflow (three-element Windkessel per outlet)

Each outlet (celiac, SMA, left/right renal, distal aorta) was coupled to a three-element Windkessel (RCR) model representing proximal (R_p) and distal (R_d) resistances in parallel with a compliance (C). The governing relations at each outlet relate the outlet flow Q, the coupled outlet pressure p, the capacitor pressure P_c, and the distal reference pressure P_d (set to 0 mmHg gauge unless otherwise noted), and are given in Appendix A. For the CFD solver, pressures were converted using 1 mmHg $= 1333.22$ dyn/cm^2 [11].

Parameter calibration (R_p, R_d, C)

For each outlet i, we first targeted the mean flow $\bar{Q}_i$ implied by the scaled inlet and the α-split above and computed the total hydraulic resistance $R_{\text{tot},i}$ from mean arterial pressure (MAP) and the distal reference pressure. We then partitioned $R_{\text{tot},i}$ into a small proximal loss and a dominant distal bed using a fraction γ (typically 0.05–0.10 for large branches, and occasionally higher for renal branches if clinically indicated). Compliance was initialized from a target diastolic time constant (e.g., $\tau \approx 0.9$–1.3 s for abdominal beds) and then tuned only via C to better match patient-specific diastolic runoff shape and late-systolic notch seen on duplex/ultrasound, while holding R_{tot} fixed to preserve mean flows. The equations used for all three calibration steps are listed in Appendix A. This tuning strategy (estimating RCR parameters from measured pressure/flow features) is consistent with contemporary boundary-condition estimation approaches [12–14].

RCR circuits were applied in SimVascular/svSolver with identical outlet assignments in pre- and post-operative models. Simulations were advanced over multiple cardiac cycles and evaluated for approach to periodicity [9, 12].

Blood properties and solver settings

Blood was modeled as an incompressible Newtonian fluid (density 1.06 g/cm^3; dynamic viscosity 0.04 g/(cm·s) $\approx$ 4 cP). Vessel walls were treated as rigid with no-slip boundary layer conditions. Transient simulations used a fixed time step of $\Delta t = 0.0002$ s and were advanced over five cardiac cycles until we achieved a stable period that minimizes relative changes between successive cycles in branch mean flows, peak systolic flows, and peak systolic pressures across all outlets. Only the final converged cycle was used for analysis.

All cases were solved with svSolver (SimVascular) using stabilized finite elements on linear tetrahedra and implicit time integration; numerical tolerances and the mesh/time discretization were held identical between each patient's pre- and post-operative models.

Endpoints and hemodynamic metrics

The primary endpoints were the pre- to post-operative changes in peak renal branch pressure (mmHg) and peak renal branch volumetric flow (mL/s), where "peak" refers to peak systole, computed from the final converged cycle under identical inflow conditions and outlet RCR parameters to isolate only the effects of each patient's newly modified flow geometry. Secondary endpoints included time-averaged pressure and volumetric flow at the renal artery outlets over the course of one cardiac cycle as well as the fractional area of abnormal time-averaged wall shear stress (TAWSS). Peak values were taken as the maximum within the last cycle. TAWSS was computed pointwise over the cycle then area-averaged per branch.

Post-operative renal branch geometry was characterized from the centerline paths extracted during SimVascular modeling. For each branch we projected the centerline onto the two-dimensional plane that best captured its overall course and located the transition point where the branch ceased running roughly parallel to the aorta and began to turn outward toward the renal ostium. This allowed us to summarize each branch by two intuitive quantities—its *tortuosity* (how much longer the branch path is than the straight line between its endpoints) and its *mean distal angle* (how

sharply, on average, the distal portion of the branch turned away from the aortic axis)—together with additional features defined in Appendix B. The associated exploratory statistical analysis is described in Appendix D.1.

Statistical analysis

Each of the 20 renal branches across the 10 patients (10 left, 10 right) was analyzed individually. For each endpoint we formed paired differences (post $-$ pre) and applied two-sided paired t-tests; we report the mean difference with 95% confidence intervals and summarize percent change as $\Delta\% = 100 \times (\text{post} - \text{pre})/\text{pre}$.

Results

Cohort and procedural characteristics

Ten consecutive patients underwent TAMBE implantation (Table 1).

DR. Tran: procedural details like target-vessel counts, device/bridge types and diameters, ballooning/relining details, and intra-operative complication rates

The mean hook angle was $40.3 \pm 15.7°$ (left $43.8 \pm 20.7°$, right $36.9 \pm 8.3°$). The mean distal branch angle from the aortic axis was $85.0 \pm 20.9°$ (left $85.8 \pm 27.7°$, right $84.1 \pm 12.4°$). Additional features extracted included axial run length, bend curvature, tortuosity, ostium cross-sectional area, and taper ratio (proximal-to-distal area ratio).

Primary outcomes: renal branch hemodynamics

Across 20 renal branches, peak renal pressure decreased from 113.1 ± 8.9 to 108.7 ± 8.5 mmHg, with a mean paired difference $\Delta = -4.39$ mmHg (95% CI -8.18 to -0.59, $p = .026$). Peak renal flow decreased from 15.55 ± 3.00 to 15.04 ± 3.37 ml/s, $\Delta = -0.50$ ml/s (95% CI -1.00 to -0.01, $p = .046$). Overall, 15 of 20 (75%) renals had $|\Delta| < 10\%$ for both peak pressure and peak

flow (Fig. 2). Five renals showed 13–22% reductions in at least one peak metric; by patient/side, these were: Patient 6 right ($\Delta P\% = -22.4\%$, $\Delta Q\% = -13.9\%$), Patient 3 left ($\Delta P\% = -8.1\%$, $\Delta Q\% = -17.8\%$), Patient 2 right ($\Delta P\% = -9.7\%$, $\Delta Q\% = -14.7\%$), Patient 6 left ($\Delta P\% = -14.6\%$, $\Delta Q\% = -0.4\%$), and Patient 1 left ($\Delta P\% = -8.9\%$, $\Delta Q\% = -13.2\%$). No branch occlusions or clinical sequelae were observed during early follow-up. Dr. Tran: median follow-up duration

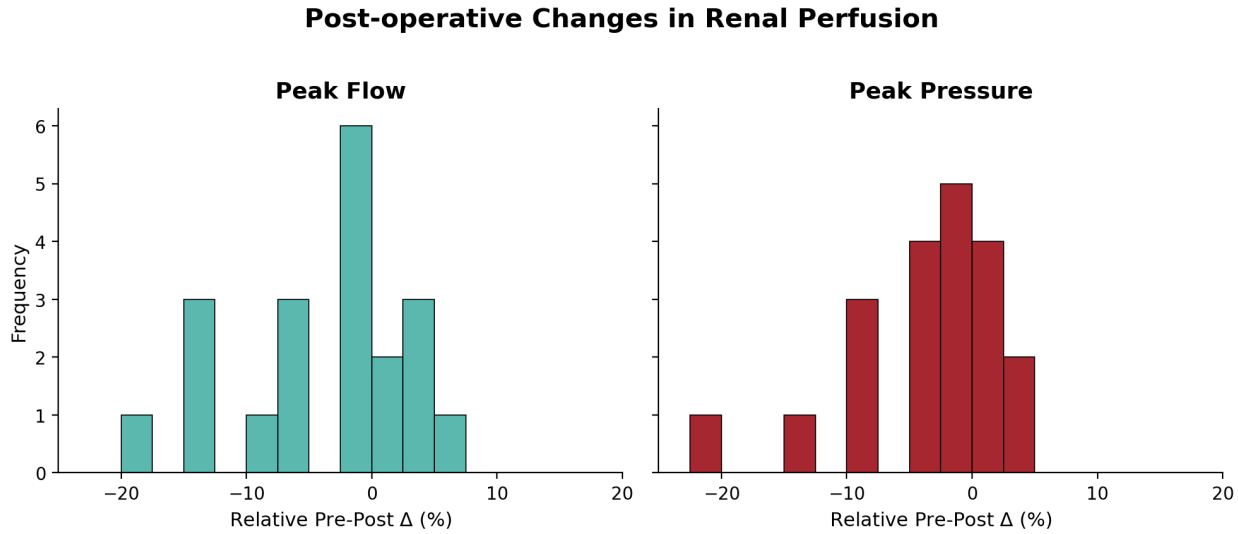


Figure 2: Distribution of post-operative changes in peak renal artery flow (left, teal) and peak renal artery pressure (right, red) across all 20 renal branches (left and right pooled). Most branches experienced less than a 10% change in either metric; five branches, visible at the left tail of each histogram, showed reductions of 13–22% in at least one peak metric.

Branch geometry and outlier hemodynamics

The five outlier branches tended to share two anatomical features: a more twisted overall course and a sharper distal turn away from the aortic axis. In an exploratory analysis, the combination of branch tortuosity and mean distal angle was the single pair of geometric features that tracked significantly with the observed reductions in peak renal flow (Appendix D.2); no single feature alone—including the naive takeoff angle at the branch origin—reached significance. Representative projected geometries for the five outlier branches are shown in Fig. 3.

Secondary outcomes

Peak celiac and SMA pressure and flow were unchanged pre- to post-operatively. The fractional area of abnormal WSS (TAWSS < 10 or > 70 dyn/cm^2) was likewise unchanged, with no shift toward low- or high-shear territory ($p = .899$).

Dr. Tran: patency rate, and reintervention count (maybe other details?)

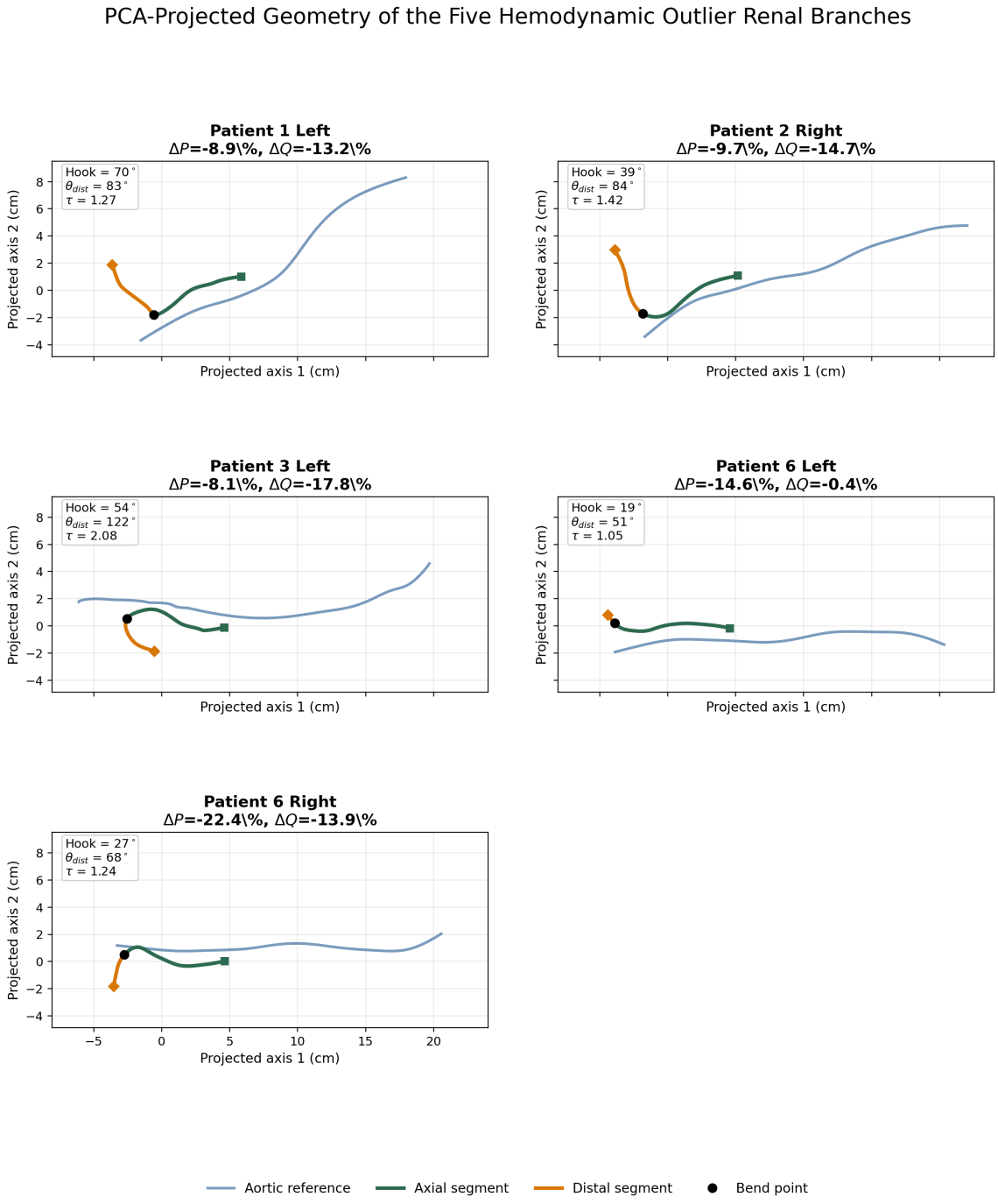


Figure 3: The five hemodynamic outlier post-operative renal branches, each projected onto the two-dimensional plane that best captures its overall course. In each panel, the adjacent aortic centerline is shown in blue, the proximal segment (running roughly parallel to the aorta) is shown in green, and the outward-turning distal segment is shown in orange; the transition point between the two is marked by a black dot. Axes are in centimeters and are displayed on common limits across panels to facilitate visual comparison of each branch's axial run, distal angulation, and overall path complexity.

Discussion

In this single-center pilot study, patient-specific CFD under identical pre/post boundary conditions showed modest mean decreases in peak renal pressure and flow after TAMBE, with three-quarters of renal branches changing $< 10\%$ and five outliers exhibiting 12–23% reductions. Celiac and SMA hemodynamics were unchanged, and the fractional area of abnormal WSS was unchanged. Importantly, no branch occlusions or clinical sequelae occurred during early follow-up. Taken together, these data suggest that, in our initial series, the off-the-shelf multibranched configuration was associated with largely preserved renovisceral hemodynamics, while specific post-operative geometry may contribute to a subtle decrease in renal perfusion in a minority of branches.

Because the TAMBE device is off-the-shelf, each branch graft exits the main body at a fixed proximal position, traverses an axial segment roughly parallel to the aorta, and then turns outward through the patient's native renal ostium. The mismatch between the fixed graft portal and the patient's native ostial position determines the axial run length, path tortuosity, and the angular deviation of the distal segment. In our exploratory analysis (Appendix D), it was the combination of branch tortuosity and distal angulation—not the hook angle or the naive takeoff angle at the branch origin—that was most closely associated with the observed reductions in peak renal flow.

Our findings align with prior f/BEVAR CFD series reporting overall preserved renovisceral perfusion under standardized boundary conditions while still detecting subtle, anatomy-linked outliers. The most recent TAMBE clinical data also demonstrate that renal branches bear a disproportionate share of target-vessel events [4]; our purely hemodynamic outliers are consistent with that signal and may help explain the clinical pattern.

Limitations include small sample size ($n = 20$ renal branches), retrospective design, rigid walls, and Newtonian rheology; and assumption-based inflow and outflow boundary conditions (e.g. not based upon velocity waveforms derived from patient duplex imaging). We also lack duplex validation in this cohort. Follow-up was short and thus we are unable to correlate our findings with longer term patency loss. The geometric analysis was performed on post-operative geometries only; the

extent to which pre-operative anatomy predicts post-operative taper remains to be established.

Future work will expand to a prospective cohort with duplex/CTA coregistration.

Conclusions

In this single-center pilot, patient-specific CFD with identical pre/post boundary conditions showed that off-the-shelf TAMBE was associated with largely preserved renovisceral hemodynamics, with only modest mean decreases in peak renal pressure and flow. No single branch geometry feature—including takeoff angle—significantly predicted these changes; instead, peak flow reductions were most strongly associated with the combination of branch tortuosity and distal angulation in an exploratory analysis, consistent with a path-complexity mechanism. These pilot data are consistent with acceptable renovisceral hemodynamics following TAMBE implantation in typical anatomy and motivate future evaluation of combined branch tortuosity and distal angulation—rather than reliance on a single angle threshold—in larger cohorts with longer clinical follow-up.

Dr. Tran: I think you might need to fill in the next four

Acknowledgments

Funding

Conflicts of Interest

Ethics Approval

Author Contributions

I can do this

Data and Code Availability

I'll handle this when everything is final; any notes about HIPAA

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Appendix A: Boundary condition parameterization

The three-element Windkessel (RCR) outlet model described in the Methods relates the outlet flow Q to the coupled outlet pressure p, capacitor pressure P_c, and distal reference pressure P_d via a mass balance on the capacitor and an algebraic outlet-pressure expression:

$$\frac{dP_c}{dt} = \frac{Q - \frac{P_c - P_d}{R_d}}{C} \tag{1}$$

$$p = P_c + R_p \cdot Q \tag{2}$$

The three RCR parameters at each outlet i were initialized by first computing a total hydraulic resistance from the target mean flow $\bar{Q}_i$,

$$R_{\text{tot},i} = \frac{\text{MAP} - P_d}{\bar{Q}_i}, \tag{3}$$

then partitioning $R_{\text{tot},i}$ into proximal and distal components,

$$R_{p,i} = \gamma \cdot R_{\text{tot},i}, \qquad R_{d,i} = (1-\gamma) \cdot R_{\text{tot},i}, \tag{4}$$

and initializing compliance from a target diastolic time constant:

$$\tau_i = R_{d,i} \cdot C_i. \tag{5}$$

Appendix B: Geometric feature definitions

For each renal branch, the branch centerline and the adjacent segment of the aortic centerline were projected onto the branch's principal-component best-fit plane (PCA; two components explaining $\geq 98\%$ of variance), reducing the analysis to two dimensions. Within this plane, the bend point b—where the branch transitions from a proximal axial segment running approximately parallel to the aorta into an outward-turning distal segment—was detected using a hybrid angle-threshold and parabola-vertex method (Appendix C). Let $\mathbf{r}_j \in \mathbb{R}^2$ denote the projected branch centerline samples ($j = 0, \ldots, N-1$), let $L = \sum_{j=0}^{N-2} \|\mathbf{r}_{j+1} - \mathbf{r}_j\|$ be the projected arc length, let $\hat{\mathbf{t}}_j$ denote the unit tangent at sample j, and let $\hat{\mathbf{t}}_{\text{ao}}$ denote the projected aortic reference direction. Three features used repeatedly in the analysis were defined as

$$\tau = \frac{L}{\|\mathbf{r}_{N-1} - \mathbf{r}_0\|}, \tag{6}$$

$$\theta_{\text{dist}} = \frac{1}{N-b} \sum_{j=b}^{N-1} \cos^{-1}\left(\hat{\mathbf{t}}_j \cdot \hat{\mathbf{t}}_{\text{ao}}\right), \tag{7}$$

$$\text{TR} = \frac{A_{\text{ostium}}}{A_{\text{distal}}}, \tag{8}$$

where τ is tortuosity, θ_{dist} is the mean distal-segment angle from the aortic axis, and TR is the taper ratio computed from proximal and distal segmentation contour areas. Additional extracted features included hook angle (angular deflection at the bend point), axial run length, axial run fraction, bend curvature, maximum angle from the aortic axis, ostium area, distal area, and a naive three-dimensional takeoff angle for comparison.

Appendix C: Bend-point detection algorithm

The bend point b for each renal branch was detected on the two-dimensional PCA-projected centerline using a hybrid algorithm with two stages and a last-resort fallback.

Primary method (angle-threshold crossing)

1. Compute the unit tangent $\hat{\mathbf{t}}_j$ at each centerline sample j via central differences.
2. For each sample, compute the angle from the aortic reference direction: $\phi_j = \cos^{-1}(\hat{\mathbf{t}}_j \cdot \hat{\mathbf{t}}_{\text{ao}})$.
3. Smooth the angle profile $\{\phi_j\}$ with a Savitzky–Golay filter (window length 15 samples, polynomial order 3).
4. Starting from 10% of the path, find the first sample j^* at which the smoothed angle exceeds a threshold of $45°$.
5. If j^* lies beyond the first 15% of the path, accept $b = j^*$.

Fallback method (rotated-parabola vertex)

If the primary method fails—for example, when the branch starts already angled relative to the aorta and never has a clearly parallel segment—the algorithm falls back to a geometric approach:

1. Rotate coordinates so that the aortic reference direction aligns with the negative y-axis.
2. Compute the smoothed gradient of the lateral (x) displacement along the path.
3. Identify the first sign change in this gradient after 10% of the path length. This sign reversal corresponds to the vertex of the branch's turning parabola—the point at which lateral deviation away from the aorta transitions to outward turning.

Last resort

If neither method produces a valid bend point, the algorithm selects the point of maximum curvature κ within the middle 70% (15%–85%) of the path.

Appendix D: Exploratory geometric regression analysis

D.1 Additional statistical methods

To explore associations between post-operative branch geometry and hemodynamic changes, we performed Pearson and Spearman correlations between each of the 12 geometric features and the percent change in peak pressure ($\Delta P\%$) and peak flow ($\Delta Q\%$). Branches were classified as hemodynamic outliers if $|\Delta P\%| > 10\%$ or $|\Delta Q\%| > 10\%$; two-sample t-tests compared each geometric feature between outlier and non-outlier groups. As an exploratory screen, all pairwise two-feature linear regression models were evaluated against $\Delta P\%$ and $\Delta Q\%$, with R^2 and F-test p-values reported. Because the geometric analysis was hypothesis-generating in a small cohort, we did not apply multiplicity correction; p-values are presented descriptively alongside effect sizes.

D.2 Geometric correlates of hemodynamic outliers

To identify which branch geometry features were associated with hemodynamic changes, we extracted 12 geometric features from each post-operative renal branch (hook angle, axial run length, bend curvature, tortuosity, distal angle from aorta, ostium area, taper ratio, and others) and correlated them with post-operative percent changes in peak pressure ($\Delta P\%$) and peak flow ($\Delta Q\%$).

No single geometric feature reached significance for predicting $\Delta P\%$ or $\Delta Q\%$ by Pearson correlation. In a two-tailed t-test comparing outlier versus non-outlier branches, the taper ratio (proximal-to-distal area ratio) trended higher in outliers (1.39 ± 0.99 vs. 0.72 ± 0.45; $p = .053$), though this did not reach statistical significance at the 0.05 level.

In an exploratory pairwise regression screen, tortuosity combined with the distal branch angle from the aortic axis emerged as the strongest joint correlate of peak flow change ($\Delta Q\%$: $R^2 = 0.31$, nominal $p = .041$). No two-feature combination reached significance for $\Delta P\%$. Neither hook angle nor the naive three-dimensional takeoff angle—measured at the branch origin—was significantly associated with either hemodynamic outcome in this cohort.

D.3 Mechanistic interpretation

Our geometric correlation analysis suggests that the mechanism underlying these hemodynamic outliers is multifactorial rather than attributable to a single geometric parameter. Notably, neither the hook angle at the branch bend point nor the naive three-dimensional takeoff angle at the branch origin correlated with hemodynamic changes. The taper ratio (proximal-to-distal area ratio) showed a trend toward higher values in outlier branches (1.39 ± 0.99 vs. 0.72 ± 0.45; $p = .053$), suggesting that luminal narrowing along the stented segment may increase resistive losses and contribute to larger pressure drops at peak systole [15], though this did not reach significance in our small cohort.

Among the exploratory pairwise models, the combination of tortuosity and distal branch angle from the aortic axis was the strongest correlate of peak flow reduction ($R^2 = 0.31$, nominal $p =$

.041). This is consistent with the hypothesis that the relevant patient-specific penalty may reflect overall path complexity rather than a single ostial angle: a more tortuous branch with greater distal angular deviation may be more likely to promote flow separation and secondary motions that depress peak instantaneous renal flow [15], even when cycle-averaged flow splits are maintained by the downstream vascular beds represented by the outlet Windkessel circuits. In other words, the clinical CFD signal appears more consistent with a multivariate geometry phenotype than with a one-parameter "takeoff angle" threshold.

can you take a look at this and edit/expand as you see fit. I want to be clear about $R^2 = 0.31$ without underselling the analysis

That said, the best two-feature model explains only 31% of the variance in $\Delta Q\%$, leaving a substantial portion unaccounted for. Several factors likely contribute to the unexplained variance. First, with only 20 renal branches and two predictors, the regression estimate itself is noisy; the true population effect size could be appreciably larger or smaller. Second, our geometric features are extracted from a two-dimensional PCA projection of each branch, which necessarily discards out-of-plane curvature and three-dimensional helical features that may influence flow. Third, the rigid-wall assumption and Newtonian rheology may obscure flow disturbances that would appear in a compliant, non-Newtonian model. Fourth, the Windkessel outlet circuits partially buffer against geometric perturbations by regulating mean flow toward the target split; this downstream compensation compresses the observable range of $\Delta Q\%$ and may mask larger instantaneous flow deficits during peak systole. Finally, luminal taper—which trended toward significance as an independent predictor ($p = .053$)—is not included in the two-feature model but may capture an additional resistive mechanism that is partially independent of tortuosity and angulation. Despite these limitations, the tortuosity–angulation signal is the strongest reproducible correlate in this cohort, and even a modest R^2 in a small pilot serves to motivate the hypothesis that multivariate path complexity, rather than a single angle threshold, drives the outlier hemodynamic response.

The absence of significant change in time-averaged WSS is consistent with this mechanism: the

flow disturbances caused by path tortuosity and angular deviation are most pronounced during peak systole, which represents only a brief portion of the cardiac cycle, and are spatially confined to the near-ostial region. When wall shear stress is averaged over the full cycle and the entire branch surface, these transient, localized effects are diluted and do not produce a measurable shift in the overall shear environment.

Taken together, these exploratory correlations suggest a plausible mechanistic explanation for why a minority of renal branches behave as hemodynamic outliers despite otherwise preserved branch performance. Branches that are both more tortuous and more angular relative to the aortic axis may incur a larger peak-systolic path-complexity penalty than branches with a shorter, smoother, more coaxial course. Because this analysis was hypothesis-generating and performed in a small post-operative cohort, we interpret these geometric associations as candidate explanatory signals rather than as a validated predictive model.